Shrey Parekh

(+91) 9004905435 | shreyparekh3@gmail.com | LinkedIn | GitHub | Portfolio | Mumbai, India

Education

SVKM's NMIMS University, MPSTME Jun 2021 – Jun 2027
Integrated B.Tech in Computer Engineering, Specialisation in AI & ML **CGPA: 3.45/4.0**
Diploma in Computer Science Engineering (2021–2024) **CGPA: 3.55/4.0**

Publications

[1] **VehicleClassAttention: Learning PCU-Weighted Signal Control for Heterogeneous Urban Traffic**,
International Conference on Computing, Communication and Networking Technologies (ICCCNet), UK, 2026, **Accepted**
→ Proposed VehicleClassAttention (VCA), a learnable attention module embedding IRC:106-1990 PCU priors into a graph attention network; 14.9% queue reduction over the strongest classical baseline.

Projects

Traffic Signal Control via Multi-Agent Reinforcement Learning Jan 2026 – Mar 2026
Python, PyTorch, torch_geometric, SUMO |

- Built a multi-agent reinforcement learning system for adaptive traffic signal control on a 9-intersection SUMO grid; a Graph Attention Network policy achieved 14.9% queue reduction and 5.7% higher throughput over Fixed-Time across 3 classical baselines.
- Designed VehicleClassAttention (VCA), a learnable attention module initialised with IRC:106-1990 PCU priors; ablation across 4 architectures over 3 seeds isolated VCA's independent 5.5% queue reduction at only 400 additional parameters.

ClassAI: Hybrid Retrieval RAG Assistant Jan 2026 – Apr 2026
Python, FastAPI, Qdrant, LangChain, HuggingFace, Ollama, Docker |

- Built a hybrid retrieval pipeline combining BAAI/bge-m3 dense embeddings (Qdrant) with BM25 sparse retrieval, fused via Reciprocal Rank Fusion and re-ranked by a BGE cross-encoder.
- Resolved catastrophic chunk fragmentation through a two-pass hierarchical splitter, reducing the chunk kill-rate from 51–75% down to 0–4% across the document corpus and recovering retrieval quality on long-form sources.
- Deployed a locally-hosted Gemma3:12b via Ollama for privacy-preserving inference with SSE streaming; hardened the API with JWT auth and rate limiting, shipped via Docker Compose.

IronLog: ML-Driven Strength Training Platform Sep 2025 – Nov 2025
FastAPI, PostgreSQL, Celery, Redis, Docker, scikit-learn |

- Engineered a 9-step asynchronous post-session analytics pipeline on Celery and Redis, combining a Gaussian Process Regressor (Matérn kernel) for 1RM prediction with PELT changepoint detection.
- Implemented a Banister fitness-fatigue ODE model (τ_{fit}=45d, τ_{fat}=15d) producing per-muscle readiness scores; fed an autoregulation engine adjusting weekly training volume from -50% to $+20\%$.

Handwriting Margin Geometry: Computer Vision Feature Extraction Sep 2024 – Dec 2024
Python, OpenCV, EasyOCR, scikit-learn |

- Built a computer vision pipeline (EasyOCR, OpenCV) to extract handwriting margin features: left-margin shape and top/bottom gradients.
- Trained multi-output SVM, Naive Bayes, and Random Forest classifiers on the extracted features; manuscript under journal review.

Experience

Data Analyst Intern, Shrey Polyplast May 2026 – Jun 2026
Python, Pandas, NumPy, Matplotlib

- Cleaned and analysed sales, dispatch, and IndiaMART enquiry data in Python and Pandas for a plastic injection-molding business, building charts that showed product-line, seasonal, and customer demand patterns.
- Used these insights to guide marketing focus and production planning across the measuring-cup and bottle-cap lines, helping prioritise high-demand products and improve production efficiency.

Skills

Languages:	Python, C, C++, JavaScript, TypeScript, SQL
ML / AI:	PyTorch, scikit-learn, NumPy, Pandas, HuggingFace, LangChain, OpenCV
Domains:	Reinforcement Learning, Graph Neural Networks (GNN, GAT), NLP, Computer Vision, RAG
MLOps & Infra:	Docker, FastAPI, Celery, Redis, PostgreSQL, Qdrant, Git

Leadership & Activities

Technical Head, IET MPSTME Student Chapter Jul 2025 – May 2026

- Built and deployed the chapter's student portal (`iet-portal.vercel.app`); ran an AI Voice Assistant workshop (50+ attendees) and a coding contest; previously Technical Executive.

Technical Executive, *ACM MPSTME Student Chapter* Jul 2023 – May 2024